

Model Risk Validation Report

Full Technical & Quantitative Analysis



GENERATED AT
2026-09-16

MONITOR ID
mon_baseline

RUN ID
run_f66c1afe...

OVERALL
HEALTH
90.1/100

HEALTHY

AI Narrative Analysis

Executive Summary

The model is healthy and stable.

Technical Summary

Mock technical summary based on findings.

Root Causes

- Mock cause 1
- Mock cause 2

Recommended Actions

- Mock action 1

Key Performance Metrics

Metric	Category	Value	Threshold	Status
Brier Skill Score	calibration	0.1165	N/A	OK
Calibration Ratio	calibration	1.0000	N/A	OK
Row Count	data_quality	5000.0000	N/A	OK
Duplicate Row Count	data_quality	0.0000	N/A	OK
Null Rates	data_quality	N/A	N/A	OK
Invalid Probability Count	data_quality	0.0000	N/A	OK
Invalid Target Count	data_quality	0.0000	N/A	OK
Monitoring Readiness Summary	data_quality	N/A	N/A	OK
Score-PD Coherence	data_quality	-0.9836	N/A	OK
Band Consistency	data_quality	1.0000	N/A	OK
DPD Buckets	delinquency	N/A	N/A	OK
Score PSI	drift	N/A	N/A	SKIPPED
ROC AUC	performance	0.7854	N/A	OK

METRIC	CATEGORY	VALUE	THRESHOLD	STATUS
Gini Coefficient	performance	0.5708	N/A	OK
KS Statistic	performance	0.4353	N/A	OK
Decile Table	performance	N/A	N/A	OK
Score Distribution Analysis	performance	N/A	N/A	OK
Approval Rate	strategy	0.8012	N/A	OK
Overall Bad Rate	strategy	0.0916	N/A	OK

Statistical Tests

TEST NAME	STATISTIC	P-VALUE	CONCLUSION	RESULT
Hosmer-Lemeshow Test	41.5414	0.0000	Reject H_0 — distributions differ	
Binomial Test: Band 600-700	225.0000	1.0000	Fail to reject H_0 — distributions are similar	
Binomial Test: Band 700-800	23.0000	1.0000	Fail to reject H_0 — distributions are similar	
Binomial Test: Band <600	210.0000	0.0000	Reject H_0 — distributions differ	
Binomial Test: Band 800+	0.0000	1.0000	Fail to reject H_0 — distributions are similar	
Mann-Whitney U: Approved vs Declined Scores	3372339.5000	0.0000	Reject H_0 — distributions differ	

Fairness Analysis (Disparate Impact)

Summary: No direct adverse impact detected.

PROTECTED GROUP	VALUE	N	APPROVAL RATE	DI RATIO
income_band	<50K	1515	76.0%	0.95
loan_purpose	home_improvement	899	78.1%	0.97
loan_purpose	debt_consolidation	1394	79.7%	0.99
home_ownership	OTHER	303	79.9%	1.00
home_ownership	OWN	612	79.9%	1.00
home_ownership	RENT	2052	80.0%	1.00
home_ownership	MORTGAGE	2033	80.3%	1.00
income_band	75-100K	994	80.6%	1.01
loan_purpose	major_purchase	909	80.8%	1.01
loan_purpose	other	708	81.1%	1.01
income_band	50-75K	1695	81.2%	1.01
loan_purpose	credit_card	1090	81.2%	1.01
income_band	100K+	796	85.0%	1.06

Vintage Analysis (Roll Rate)

Note: True roll rate matrix requires two time-snapshot datasets. This shows current DPD bucket distribution.

Severe Delinquency Rate: 6.60%

BUCKET	COUNT	PERCENTAGE	CUMULATIVE %
Current (0 DPD)	4250	85.0%	85.0%
1-29 DPD	0	0.0%	85.0%
30-59 DPD	215	4.3%	89.3%
60-89 DPD	205	4.1%	93.4%
90+ DPD	330	6.6%	100.0%

Override Analysis

"Overridden accounts show a HIGHER bad rate than model-approved accounts (31.0% vs 6.1%), indicating possible excessive risk taking on model-declines."

MODEL DECLINES 172	OVERRIDES 29
OVERRIDE RATE 16.9%	OVERRIDE BAD RATE 31.0%

Conclusion: Overrides are performing worse than standard model approvals.

"Overridden accounts show a HIGHER bad rate than model-approved accounts (22.7% vs 5.1%), indicating possible excessive risk taking on model-declines."

MODEL DECLINES 757	OVERRIDES 269
OVERRIDE RATE 35.5%	OVERRIDE BAD RATE 22.7%

Conclusion: Overrides are performing worse than standard model approvals.

"Overridden accounts show a HIGHER bad rate than model-approved accounts (13.0% vs 3.1%), indicating possible excessive risk taking on model-declines."

MODEL DECLINES	OVERRIDES
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2115

1291

OVERRIDE RATE

61.0%

OVERRIDE BAD RATE

13.0%

Conclusion: Overrides are performing worse than standard model approvals.

"Overridden accounts show a HIGHER bad rate than model-approved accounts (8.3% vs 1.8%), indicating possible excessive risk taking on model-declines."

MODEL DECLINES

3746

OVERRIDES

2771

OVERRIDE RATE

74.0%

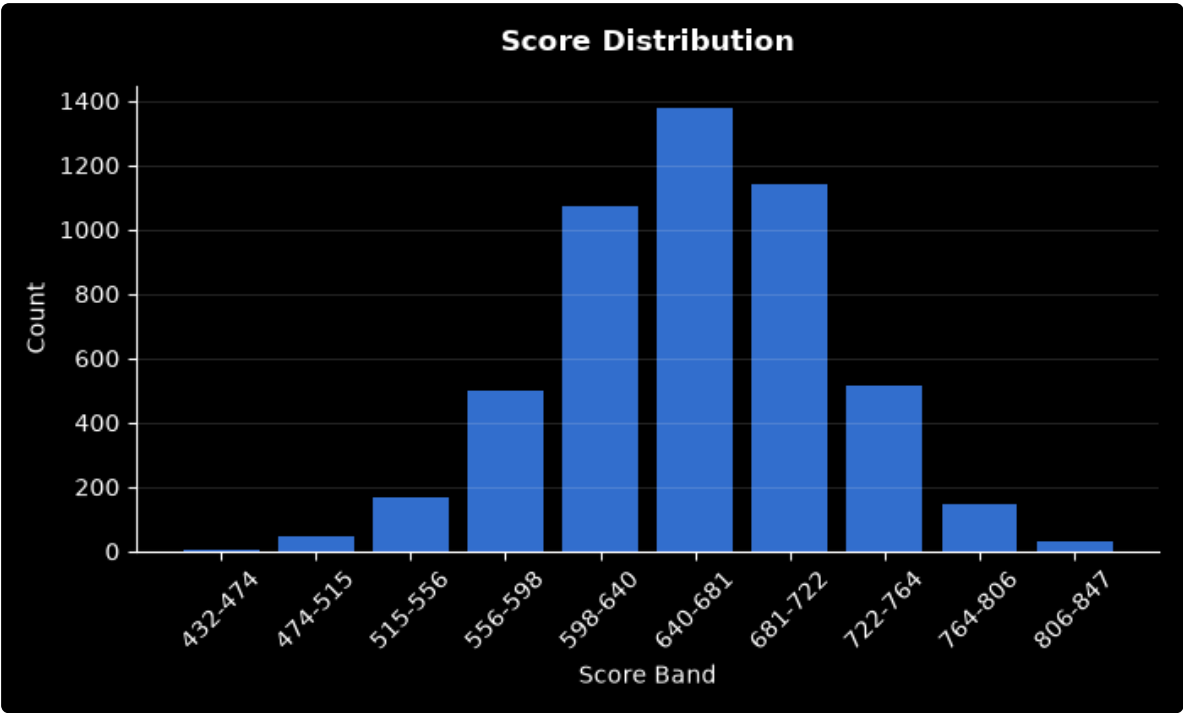
OVERRIDE BAD RATE

8.3%

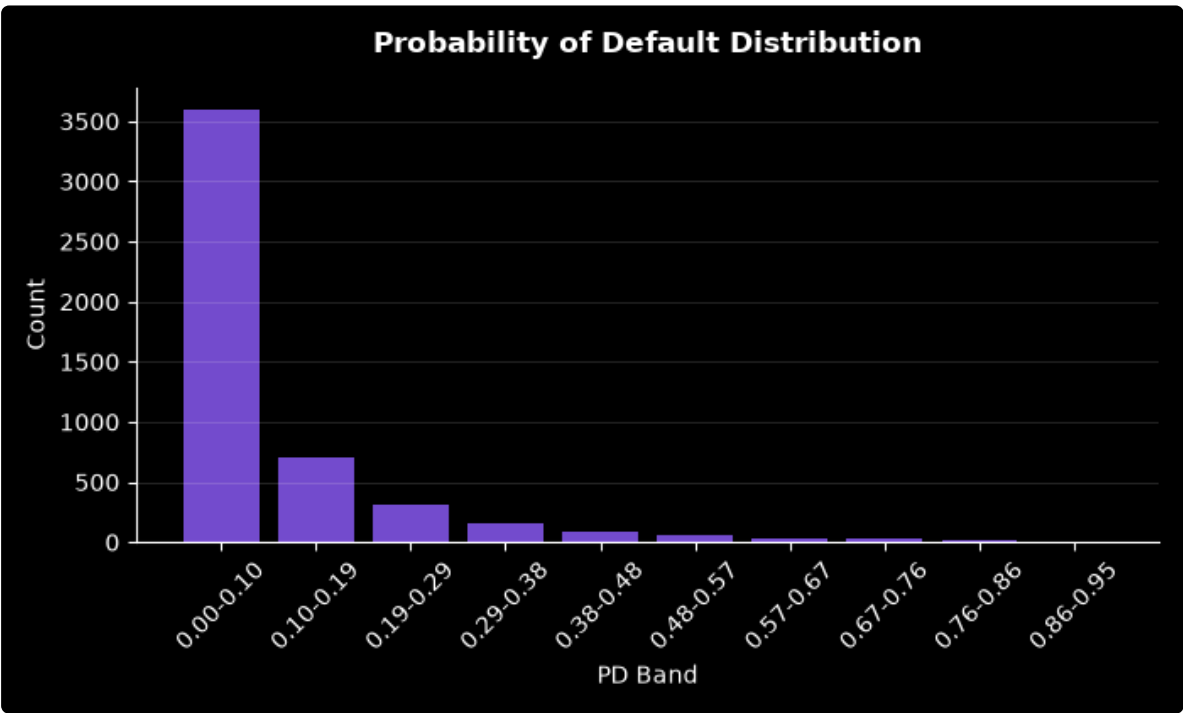
Conclusion: Overrides are performing worse than standard model approvals.

Model Visualizations

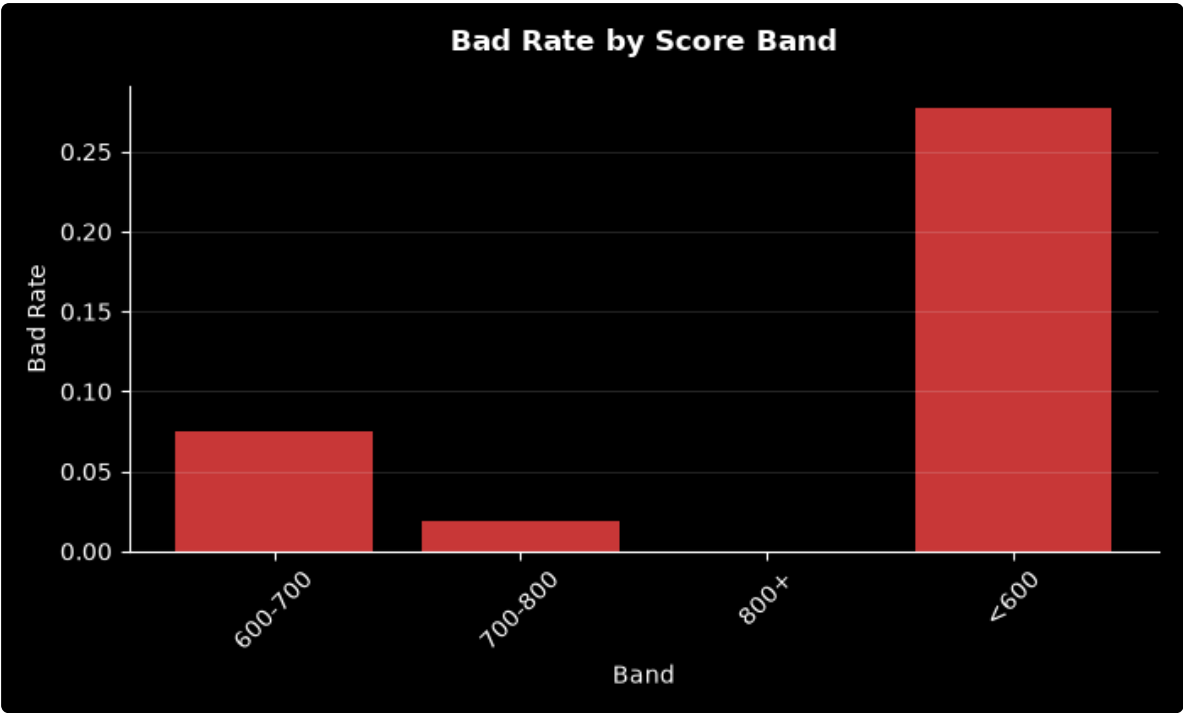
CHART_SCORE_DISTRIBUTION



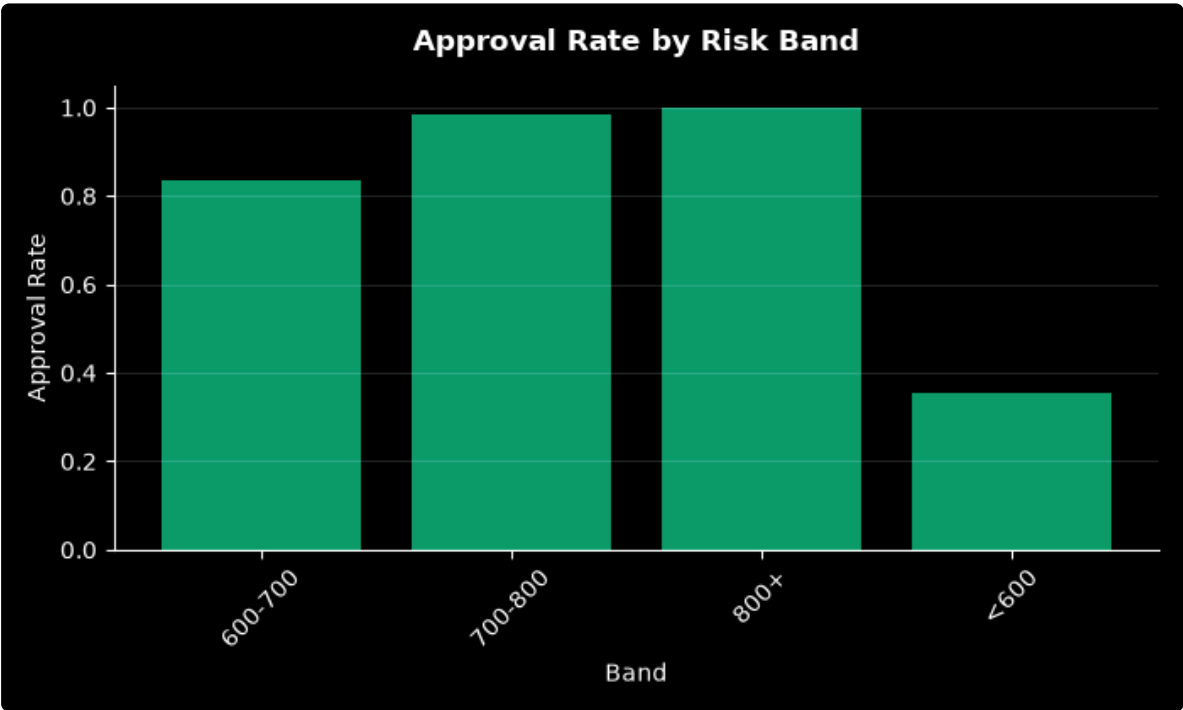
CHART_PD_DISTRIBUTION



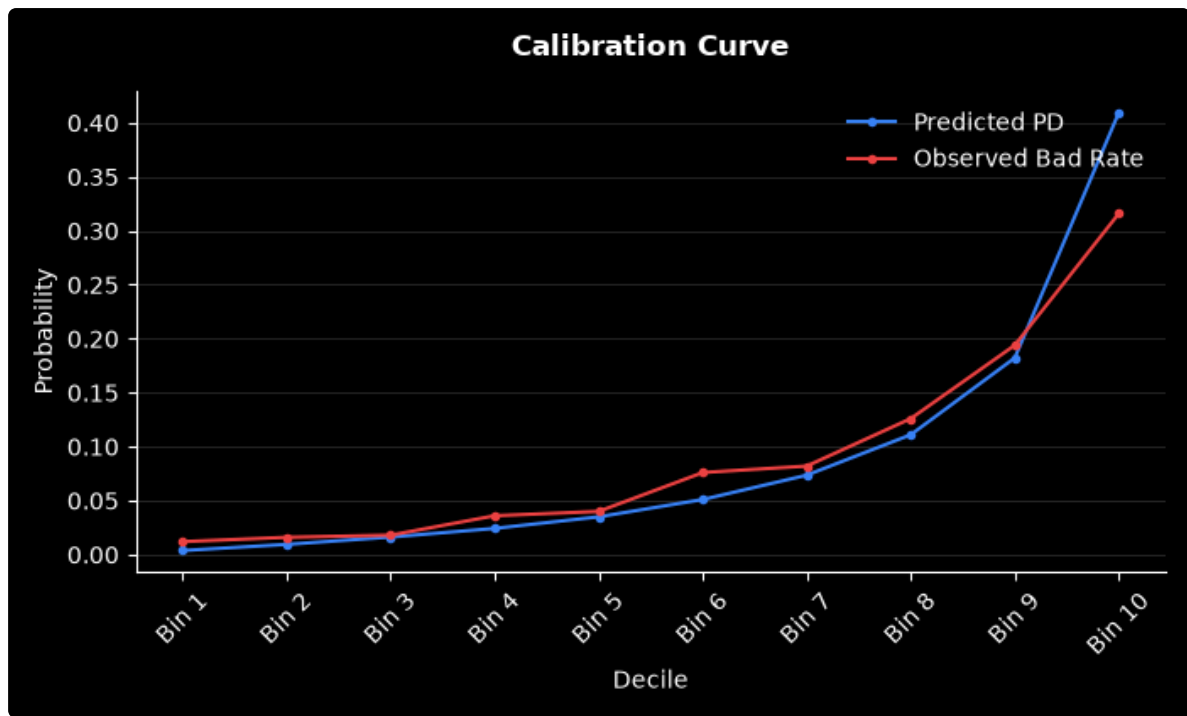
CHART_BAD_RATE_BY_SCORE_BAND



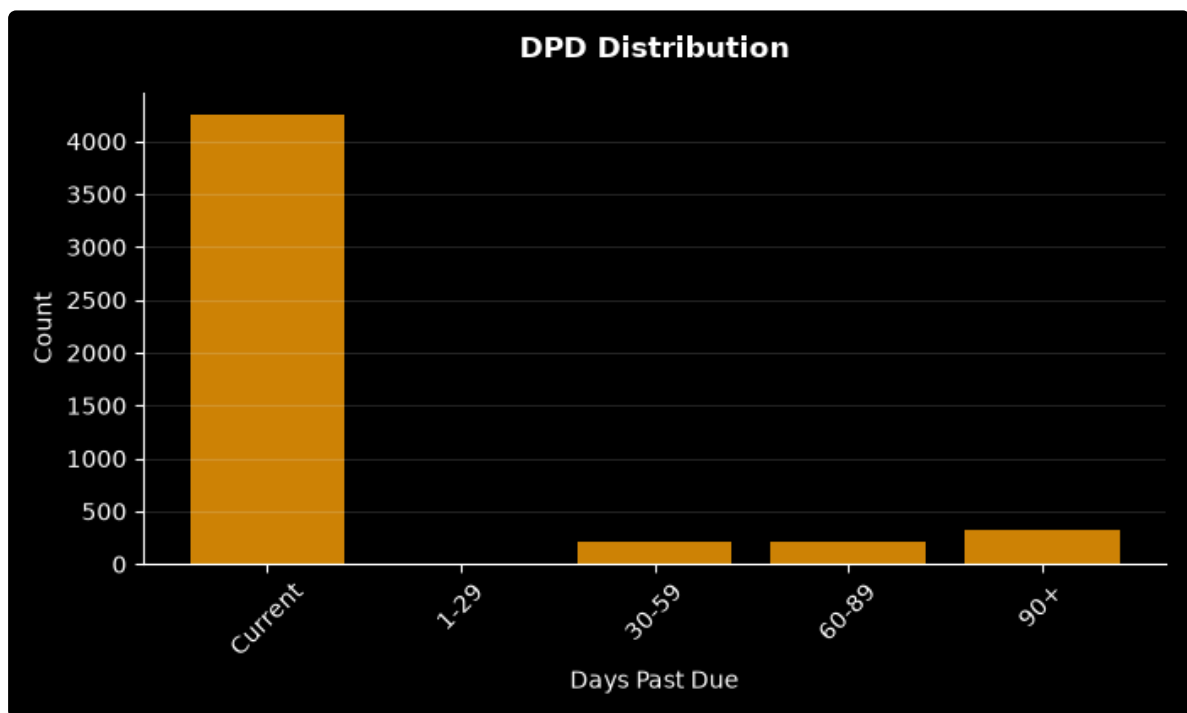
CHART_APPROVAL_BY_RISK_BAND



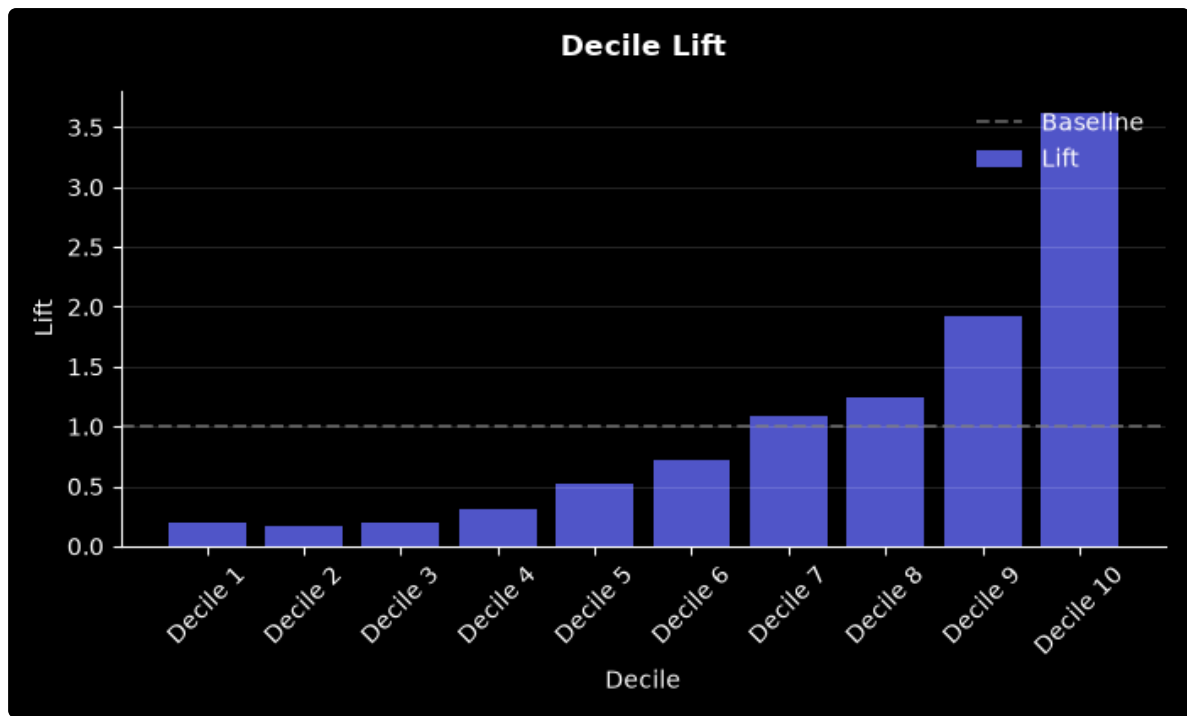
CHART_CALIBRATION



CHART_DPD_DISTRIBUTION



CHART_DECILE_LIFT



CHART_ROC_CURVE

